

AI-Driven Lead Management Automation

Walkthrough

Environment setup, workflow execution, and edge-case handling

Project	AI Lead Automation (FastAPI + Gemini + Odoo)
Stack	Python, FastAPI, Google Gemini, Gmail SMTP, Odoo CRM
Test mode	21/21 tests passing, fully offline (mock mode)

1 Overview

This service automates inbound lead handling for a WordPress contact form, up to (but not including) quotation generation. On form submission it classifies the inquiry with Google Gemini, generates tailored follow-up questions, emails the prospect a contextual reply with the Company Profile PDF attached, and logs the lead into Odoo CRM with department routing - with zero manual triage.

2 Environment Setup

Steps to run the service locally:

1. `pip install -r requirements.txt`
2. Create a `.env` file with:
`WEBHOOK_SECRET, GEMINI_API_KEY, GMAIL_ADDRESS, GMAIL_APP_PASSWORD,`
`ODOO_URL, ODOO_DB, ODOO_USERNAME, ODOO_API_KEY`
`LLM_MODE / EMAIL_MODE / ODOO_MODE = mock (default) or live`
3. `uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload`
4. `pytest -v` (21 tests, fully offline, zero API calls)

In mock mode (the default), the app runs end-to-end with zero external API calls - safe for development and grading without spending real Gemini/Gmail/Odoo credits. Setting a mode to 'live' in `.env` switches only that integration to the real service; the webhook contract and response shape never change.

3 Workflow Execution

A single request to `POST /webhook/lead` flows through the code in this order:

- 1 app/main.py**
Receives the webhook and checks the `X-Webhook-Secret` header (401 if wrong).
- 2 app/schemas.py**
`InboundForm` validates name / email / message via `Pydantic`.
- 3 app/services/prefilter.py**
`normalize()` cleans whitespace; `is_junk()` drops empty/garbage input.
- 4 app/services/llm.py**
`generate_ai_result()` calls Gemini (or the mock heuristic) in a single call for intent, category, confidence, summary, and 3-6 follow-up questions.
- 5 app/main.py**
Returns the classification response immediately, then queues two background tasks so the caller never waits on email or CRM.
- 6 app/services/emailer.py**
`send_lead_email()` emails the prospect, attaching `Company_Profile.pdf` and the AI-generated questions.
- 7 app/services/odoo.py**
`log_lead()` creates a `crm.lead` record in Odoo, routed to the correct Sales Team (`team_id`) for the department, with phone carried from the payload and priority set from the AI confidence.

Example: detailed lead

```
POST /webhook/lead
Header: X-Webhook-Secret: <secret>
{
  "name": "Jane Cooper", "email": "customer@acme.com",
  "message": "We want an e-commerce website to sell about 200
              products with Stripe payments and a blog."
}

-> 200 { "status": "accepted", "department": "Web",
        "category": "web_development", "summary": "...",
        "questions": [ ...3-6 tailored questions... ] }
```

4 Edge Cases Handled

Vague or empty input

A near-empty message (e.g. only whitespace) is rejected by `is_junk()` before any API call is made, returning `{"status": "ignored"}`. A message that is too short or vague to classify confidently is still sent to the model, which is instructed (via few-shot examples in the prompt) to return the `'needs_clarification'` category with broad, safe follow-up questions rather than guessing.

Invalid or missing webhook secret

Every request must include a matching `X-Webhook-Secret` header, checked against `WEBHOOK_SECRET` from `.env`. A missing or wrong header returns 401 before any other processing happens.

Gemini failures (rate limit, timeout, malformed response)

Transient errors (rate limit, timeout, service unavailable) are retried up to 3 times with exponential backoff via `tenacity`. If Gemini still fails, or returns a response that isn't valid JSON, `generate_ai_result()` catches the error and falls back to `fallback_result()` - a complete, valid `AIResult` in the `'needs_clarification'` category. The customer still receives a coherent reply and the lead is still logged, just flagged for human review. The flow never hard-fails on an LLM error.

Odoo / email delivery failures

Email and CRM logging run as decoupled FastAPI background tasks, so a slow or failing integration on one side never blocks the other or the webhook response. If the Odoo API call fails after its own retry/backoff, the lead payload is appended to `failed_leads.jsonl` as a dead-letter queue instead of being silently lost.

5 Testing

`pytest -v` runs 21 tests covering prefilter, schema validation, LLM parsing/fallback, email building, Odoo payload building (team routing, phone, priority), and a full integration test of the webhook in mock mode - all fully offline with zero API calls.